LITERATURE STUDY FINAL PAPER- CPRE 5480

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"TSN's Application in Industrial Automation, Robotics and Automobiles"

Problem Statement:

Modern industrial and vehicular systems depend on fast, predictable communication. In these environments, even a single delayed or lost message can stall a production line, interrupt robotic movement, or risk the safety of passengers and operators. With more than 6 million vehicle accidents each year in the United States, the role of deterministic networking in safety-critical systems is substantial.

To support this, cyber-physical systems (CPS), such as industrial automation equipment, collaborative robots, and automotive platforms, require communication that is not only fast, but reliable, deterministic, and with low latency. Wired Time-Sensitive Networking (TSN) standards, including IEEE 802.1Qbv (for scheduled traffic) and IEEE 802.1AS (for microsecond-level synchronization), already provide these guarantees in controlled environments. They work by dividing time into precise windows, reserving bandwidth, and synchronizing clocks across the network—ensuring that critical messages arrive on time.

But the real world isn't always wired. While the 2020 revision of IEEE 802.1AS introduces wireless extensions, wireless TSN still lags behind the predictability of wired deployments. Why? Wireless links operate in a shared, often congested spectrum where interference, signal fading, and physical obstacles are everyday challenges. These factors directly undermine the tight timing and reliability requirements that safety-critical systems demand.

In wireless networks, propagation delay and jitter make it hard to maintain the microsecond-level guarantees that TSN requires. Interference can delay or drop packets, disrupting the carefully orchestrated schedules that keep robots coordinated and vehicles safe. Even common wireless reliability techniques—such as retransmissions and random backoffs—introduce variability that clashes with the stringent real-time demands of safety-critical systems.

This literature study explores how emerging wireless TSN and MAC-layer adaptations aim to bring wired-like determinism to Wi-Fi, 5G, and hybrid networks. We focus especially on two often-overlooked but essential areas: admission control (can we safely add new traffic?) and network planning (how do we design for determinism at scale?).

Introduction

Industry 4.0 has transformed factories, warehouses, and roads into interconnected ecosystems where sensors, robots, and vehicles must communicate in real time. In these settings, deterministic communication is non-negotiable—a dropped or delayed packet can mean a stalled assembly line, a robotic collision, or a vehicular malfunction.

Wired TSN delivers this determinism through a suite of well-defined standards: traffic shaping to prioritize critical data, scheduled transmission windows to avoid collisions, and network-wide clock synchronization to coordinate actions down to the microsecond. These mechanisms enable safety-critical systems to operate with predictable, ultra-low latency currently.

Yet, as CPS grows more mobile and flexible, wireless and hybrid architectures are becoming the new norm. Wireless offers clear benefits: mobility, easier installation, and reduced cabling. But it also introduces major challenges: the unlicensed spectrum is crowded and unpredictable; signals fade with distance, and interference from other devices is common. These uncertainties make it difficult to directly transfer TSN's wired techniques into wireless environments.

Recent research shows promise. The PktR (predictable per-packet communication reliability) framework, proposed by Meng and Zhang [1], combines scheduling and dynamic power control in 5G networks to achieve ultra-reliable low-latency communication (URLLC). By adapting transmissions in real time, PktR maintains strong signal quality even under variable channel conditions. This illustrates a key insight: wireless determinism is possible, but only if the network can actively manage interference and variability—a core challenge for wireless TSN.

In this study, we review current solutions, identify open problems, and highlight promising avenues for future research. We pay special attention to cross-domain scheduling (coordinating wired and wireless segments), intelligent admission control (using SDN and ML for real-time decisions), and the tools and testbeds needed to evaluate wireless TSN in practice.

Cyber-physical systems networking problems:

Cyber-physical systems rely on a real-time exchange of sensor data, control commands, and status updates. When communication fails to meet strict timing requirements, performance may degrade. In safety-critical settings, the stakes are even higher. For example, in collaborative robotics or autonomous vehicle platoons, even a small delay can trigger a chain reaction: a robot might misalign a part, or a vehicle might brake too late, leading to a crash.

For CPS, deterministic networking means delivering three key guarantees:

- Bounded latency: messages must arrive within a known time window.
- Minimal jitter: variation in delay must be tightly controlled.
- Very low packet loss: packet drop must be minimized, especially for critical commands.

Achieving these in wireless networks is inherently difficult. Unlike wired Ethernet, where the medium is controlled and collisions are avoided, wireless links are shared, unpredictable, and

vulnerable to external interference. Mobility, multi-hop routing, and fluctuating channel conditions add further complexity, making it hard to directly apply TSN's collision-free scheduling models.

Two problems are particularly central to challenges to Cyber Physical Systems:
1. Time Synchronization (IEEE 802.1AS/gPTP)

Precise synchronization is the foundation of TSN. It ensures that all devices share a common timeline, allowing scheduled transmissions to align. In wireless networks, clock drift, asymmetric delays, and packet loss can desynchronize nodes, causing schedules to misalign and real-time guarantees to break down.

2. Admission Control and Resource Mapping

In wired TSN, reserving bandwidth is straightforward. In wireless, resources must be dynamically mapped to TSN streams while accounting for retransmissions, rate adaptation, and interference. Admission control must evaluate not only bandwidth, but also worst-case latency and interference scenarios, before allowing a new stream.

A compelling example is provided by Cyber-Physical Scheduling (CPS) for vehicular networks (Li et al., 2020), which utilizes geometric interference modeling to predict and coordinate transmissions between vehicles. This shows that with careful modeling and scheduling, strong reliability is possible even in dynamic wireless environments—a principle that wireless TSN can build on.

Because many CPS are safety-critical, both offline network planning (designing schedules, selecting channels, and adding redundancy) and online admission control (making real-time flow decisions) are essential to maintain determinism amid wireless uncertainty.

Current TSN Applications and Challenges:

TSN has been widely adopted to enable deterministic Ethernet communication in industrial automation, robotics, and automotive domains. Standards, such as IEEE 802.1Qbv (Time-Aware Scheduling) and IEEE 802.1Qcc (Stream Reservation), have been leveraged for vehicle and robotics applications.

In the realm of robotics, experiments have been conducted to demonstrate the inefficiencies of traditional Ethernet-based solutions. In [5], researchers created a simulated robotics network that replicated an environment that included 2 actuators, a sensor, and a robotic arm on a shared, high-bandwidth network. During the experimentation, researchers examined how traditional Ethernet-based solutions compared in performance to TSN time-aware shaping procedures. The results of the experimentation concluded that time-aware shaping and TSN were able to provide

consistent isolation of priority traffic and low jitter, even in environments with high-bandwidth traffic. From the author's work, the apparent benefits of using TSN in robotic applications are made clear, and the author's work highlights how future expansions in industrial robotics will require more in-depth and comprehensive TSN solutions.

Examining the current applications for TSN in the automotive industry, it is readily apparent that the performance capabilities provided by TSN enable deterministic performance not only within a single vehicle but also across multiple vehicles platooned together and communicating vehicle-to-vehicle (V2V). The expansion of the integrated vehicle system to include other vehicles and smart infrastructure highlights the need for ultra-reliable traffic transmission. Without deterministic network performance, key functions of vehicle automation, such as collision detection and avoidance, could be ineffective, resulting in injury or death to operators or passengers.

While the use cases of TSN in automotive and robotic network settings are clear, the challenges present to both operating environments may be less apparent. When working with robotics automation networks, it is important to consider the operating environment of these systems. Robotics networks will often function in environments with high network noise and interference. Factory floors or remote manufacturing locations provide environments that can limit or inhibit reliable network performance. Future work is needed to address and accommodate these potential limitations to TSN in these robotic use cases as vehicle platooning capabilities expand and smart infrastructure becomes more prevalent.

Similar to the issues faced by robotics networks, automobile TSN networks face many challenges related to the operational use cases of these technologies. The operating environment for automotives can range from densely populated cities to remote and rural locations. Operating in city environments can be challenging, as the city's infrastructure may interfere with the performance of certain networked elements. Additionally, the prevalence of multiple vehicles within a V2V network could create dynamic network topology problems.

In both robotic and vehicular implementations of TSN, the core issue that remains is the problem of dynamic environments. A shifting environment produces varying levels of interference that can contribute to adverse network performance. These unique network characteristics highlight the need for future TSN implementations to not only be scalable in their capabilities but also robust.

MAC Protocols in TSN Applications:

For various TSN applications, there are corresponding MAC protocols that best fit each use case. Additionally, there is still a lot of research that needs to be done in the field of MAC protocols before any one protocol can be identified and isolated as the most optimal for TSN applications in either a robotics or an automotive setting.

Examining MAC protocols in the robotics use case for TSN, it can be observed that the Time-Aware Scheduler (TAS), based on a TDMA defined in 802.1Qbv, guarantees access to the MAC is made without delay in time-critical applications [5]. Through these behavioral characteristics, the performance benefits of the identified MAC protocol at the MAC layer are apparent. Future identification of alternative MAC protocols could provide additional options; however, the research outlined in [5] indicates that the current solution, which utilizes TAS TDMA in 802.1 Qbv, provides ample performance benefits for robotics use cases.

From the perspective of vehicular TSN applications, the use case and scope of MAC protocols can be more expansive, and their performance aspects can vary accordingly. These protocols are partially identified in [2]. First and foremost, one of the most important MAC protocols highlighted is the IEEE 802.11p standard. 802.11 p utilizes CSMA/CA to coordinate access and control interference. The paper [2] also highlights that this protocol is already in use for field deployments from the United States Department of Transportation (USDOT). Additional MAC protocols are identified in the paper, namely the ADHOC MAC (AMAC) protocol and the VDDCP MAC protocol.

The AMAC protocol is designed to leverage TDMA for slot-based reservation. With this protocol implementation, vehicles can transmit the slots they have uniquely reserved without worrying about carrier sensing [2]. VDDCP is a protocol that utilizes time slots for different roads, where vehicles compete with each other for channel access. VDDCP leverages many of the same features provided by TDMA slot-based reservation as AMAC [2]. The core of VDDCP is based on slot reservation, which stems from the protocol being a TDMA-based system.

As highlighted above, there are various approaches for MAC protocols, each with its own use case and tradeoffs. By understanding and analyzing these protocols, the most suitable and applicable MAC layer solutions can be identified and applied in various scenarios to achieve maximum performance gains.

Existing Solutions:

Existing TSN solutions focus primarily on scheduling, synchronization, and traffic shaping. Key examples include: IEEE 802.1Qbv Time-Aware Shaper for deterministic scheduling, IEEE 802.1Qcc Stream Reservation Protocol (SRP) for resource reservation, and IEEE 802.1AS for precise time synchronization across devices.

In an industrial automation, robotics, and automobile environment, one of the key TSN solutions implemented is IEEE 802.1AS, which allows for time synchronization across all devices in these environments[9]. This ensures that all devices within the TSN network share a common time reference, which is essential for coordinating actions and facilitating low-latency data delivery. In an automation environment, this is critical since the systems are constrained by stringent timing requirements. Deploying TSN enables meeting the high-bandwidth and low-latency requirements for vehicle communication[7]. Similarly, in a robotics and industrial environment, time-sensitive communication needs to be effectively delivered in order to ensure that all devices carry out their tasks correctly. Another key solution is IEEE 802.1Qbv Time-Aware Shaper. This solution is based on TDMA, as it divides the cycle time into time slots dedicated to different classes of services[7]. This solution utilizes gates for each queue, allowing the gate to open and close the transmission of that queue[7]. This solution is also crucial in all environments, as it enables critical traffic to access the medium without delay. This is extremely important to ensure devices operate as specified and transmit and receive communication at their designated times. Lastly, a key solution for TSN is IEEE 802.1Qcc Stream Reservation Protocol (SRP)[7]. This solution is critical in industrial, automation, and robotics since it allows supporting more streams and supports layer-3 stream reservation[7]. This ensures that specific data streams receive the necessary bandwidth, and timely delivery is achieved. Ensuring timely delivery of time-sensitive communication in critical industrial, robotics, and automation environments.

Beyond these mechanisms, admission control and network planning are critical to maintaining deterministic behavior as systems scale.

TSN solutions for admission control utilize network resource reservation to ensure timely and reliable data delivery. Solutions focus on centralized and distributed architectures, utilizing a combination of configuration standards and traffic shaping mechanisms, such as credit-based shaper (CBS) and asynchronous traffic shaping (ATS)[7]. IEEE 802.1Qav is a traffic shaping mechanism that regulates the transmission of time-sensitive traffic to ensure it does not exceed its reserved bandwidth, utilizing a credit system [7]. The IEEE 802.1Qbv Time-Aware Shaper utilizes scheduled traffic to open and close gates for different traffic classes at predetermined times[7]. The architecture for admission control is outlined in the IEEE 802.1Qcc Stream Reservation Protocol (SRP)[7]. This is a fully centralized approach that utilizes Centralized Network Configuration (CNC) to discover the network topology, capabilities, and resources [7].

The CNC calculates routes and schedules and then pushes configurations to all network devices. Using a combination of these IEEE standards allows TSN to achieve admission control.

Focuses on designing and scheduling the network to meet deterministic, real-time communication requirements by considering bandwidth, time synchronization, and low latency. A key component of TSN network planning includes requirements analysis, which is identifying and defining the bandwidth, low latency, and precise time synchronization needs for applications. Traffic management is also an aspect used in network planning. This includes traffic shaping, priority-based scheduling. Scheduling is also important in network planning, allowing real-time traffic to be transmitted and received within the defined timings.

Open Problems :

An open challenge in TSN includes MAC protocols in TSN applications. End-nodes may struggle to manage and learn multiple MAC addresses efficiently. Especially when multiple time-sensitive flows are involved, which can lead to unreliability or network management complexities. Since TSN is a collection of protocols, manufacturers can customize their implementation. Making sure interoperability at the MAC and physical layers between different vendor hardware can be challenging.[6] Some challenges revolve around diversity in data formats and standards. Systems utilize various data formats and may require specialized middleware or conversion tools. Another challenge for interoperability includes ensuring real-time data processing and analysis. Data is being collected from multiple sources, and it must be processed and analyzed to ensure real-time deadlines. Lastly, a challenge to interoperability arises from the varied communication protocols used by multiple CPS devices. The overall functionality of the CPS is largely determined by effective communication; however, due to the different protocols involved, this can be a significant challenge.[11] Additionally, some security features may also cause MAC issues in TSN. Some frames, such as Type Length Value (TLV) frames used for management in time synchronization protocols, can be used by attackers to disrupt time synchronization at the MAC level.[6]

One challenge in TSN includes maintaining time synchronization across all network devices when applying the master-slave-based Generalized Precision Time Protocol (gPTP)[12]. Synchronization errors can occur, including time value error, incorrect time-related information, and asymmetry network delay.[7] Security attacks can also occur where compromised devices in the synchronization spanning tree propagate erroneous time information, which can lead to accumulated errors and synchronization failures. However, to address these time synchronization problems, IEEE 802.1AS only provides limited fault tolerance, relying primarily on the Best Master Clock Algorithm (BMCA)[12]. BMCA offers basic redundancy by continuously evaluating all potential Grandmaster clocks in the network and selecting the one with the highest priority.[7] When the active GM fails or becomes unavailable, BMCA triggers a reselection process and transitions the network to a new GM[12].

Another challenge in TSN is in the IEEE 802.1Qbv Time-Aware-Shaper, and that is trying to figure out when to open and close the gates[7]. This complexity arises due to the NP-completeness of the TSN scheduling problem so no polynomial time scheduling algorithm exists unless P=NP[7]. The primary challenge of creating TAS-based schedulers is balancing the trade-off between efficiency and precision.[7] This trade-off arises from two main considerations: the choice of scheduling model, such as to allow flow preemption, frame fragmentation or to generate the schedule and routing path jointly[7]. Using a complex scheduling model allows for complex schedulability but also increases the computational overhead, which is not ideal for resource-constrained systems. This challenge affects the latency guarantee.[7]

Additionally, IEEE 802.1Qbu Frame Preemption is beneficial for achieving bounded low latency, especially for critical traffic, by preempting the transmission of non-critical traffic. However, this also poses a challenge for latency guarantees [7]. This standard only defines a one-level frame preemption system that classifies traffic as either express (high-priority) or preemptable (low-priority). While this ensures that critical traffic can preempt lower-priority frames, it lacks flexibility because frames within the same category cannot preempt one another. Therefore, there might be a higher-priority frame within the same priority class.

TSN enhances industrial network reliability through standards such as IEEE 802.1CB, 802.1Qca, and 802.1Qci; however, their lack of detailed implementation guidance leaves open challenges.[7] Reliability techniques rely on space redundancy, which creates multiple paths and duplicate frames to handle permanent faults, but this approach consumes significant bandwidth and increases scheduling complexity [9]. Time redundancy, on the other hand, retransmits frames to address transient faults more efficiently by adapting retransmission counts to link fault probabilities[9]. Resource-sharing approaches can reduce overhead, but they require a complex analysis of transmission success probabilities.[7]

In a TSN network, time itself can be used as an attack vector. A missed deadline during critical communication transmission, due to an unexpected delay induced by an attacker, can have severe consequences. Therefore, latency is not only the primary performance metric but also a safety and security issue. Aspects of TSN, such as time synchronization, scheduling of data traffic, and orchestration and reservation of network resources, can be used by attackers to hinder deterministic communications. Ensuring the security of TSN should be a prior design concern.

Promising Solution Approaches:

A possible solution to address the challenge of MAC protocols in TSN is streamlined address use. Networks should aim to limit TSN bridges to using one globally unique MAC address for end-node communications. Additionally, employing real-time capable firewalls with standardized security protocols for port authentication and integrity can help mitigate security threats to time synchronization.

To solve the problem of maintaining time synchronization among all devices within a network. IEEE 802.1ASdm helps enhance resilience to synchronization failures. This protocol defines a hot standby mechanism to maintain two time domains simultaneously, without relying on BCMA [12]. This standard enables the GM to seamlessly take over if the primary clock fails. This provides fault-tolerant and fast recovery; additionally, it helps minimize local clock drift, which is crucial for maintaining high accuracy in industrial settings[12]. This standard helps strengthen TSN's fault-tolerance and increase timing guarantees for applications.

A possible solution for when to open and close the gates in the IEEE 802.1Qbv Time-Aware-Shaper standard is to use Integer Linear Programming (ILP). IEEE 802.1Qbv utilizes a Gate Control List (GCL), a pre-configured, time-based schedule, to determine when to open and close the gates[9]. Each gate on the bridge's egress port is controlled by a GCL, and the list contains entries specifying the state for each gate at different time slots within the cycle. ILP is a mathematical approach that formulates the problem as a set of equations and inequalities to find the optimal GCL[9]. Heuristics and algorithms, such as subproblem divisions and genetic algorithms, can also be used[9].

IEEE 802.1Qbu's inability to preempt frames within the same class could be addressed by implementing multi-level preemption[7]. Preemption enhancements to the 802.1Qbu standard allows for the creation of more than two priority classes. With added preemption capabilities, higher-priority frames to preempt lower-priority frames even within the same initial category. This is done by creating multiple express classes rather than just one. Another approach to addressing this preemption issue within the same priority class is to combine preemption with other traffic shaping mechanisms, such as IEEE 802.1Qav [7].

The lack of reliability in IEEE 802.1CB, 802.1Qca, and 802.1Qci stems from the lack of definition of all necessary parameters in the standards[7]. A solution to this is tuning configuration parameters, such as sequence history, recovery algorithm choice, and burst size. Additionally, improving algorithms to select more reliable paths can help enhance reliability [9]. Another possible solution to ensuring redundancy without increasing bandwidth is the implementation of HSR-aware TSN[13]. This is high availability seamless redundancy that,

when implemented with TSN, achieves redundancy with the TSN guarantees of latency and determinism[13].

A possible solution to the time attack vector is the use of a machine learning (ML)-based intrusion detection framework against low-rate denial-of-service attacks on TSN-based platforms[10]. Intrusion detection systems play a critical role in defending TSN systems. Research in this area includes creating a dataset to analyze the impact of malicious timing behavior of TSN streams[10]. Using the dataset, a machine learning-based intrusion detection model, which enables the assessment of an ML algorithm in terms of attack detection performance and computation cost[10].

Conclusion:

Time-Sensitive Networking has proven to be a reliable solution for delivering deterministic communication in wired cyber-physical systems. Standards such as IEEE 802.1Qbv, IEEE 802.1AS, and IEEE 802.1Qcc provide the core capabilities needed for real-time operation, including precise time synchronization, scheduled traffic, and guaranteed resource reservation. These mechanisms have enabled dependable performance in industrial automation, robotics, and automotive systems where timing failures can lead to safety risks or mission failure.

However, extending TSN beyond wired environments introduces significant challenges. Wireless links operate in shared and unpredictable conditions where interference, mobility, and changing network topology directly impact latency and reliability. As discussed throughout this study, these conditions make it difficult to maintain the strict timing guarantees that cyber-physical systems require. While current MAC-layer approaches such as TDMA-based scheduling offer partial solutions, they do not fully address the complexity of wireless determinism on their own.

This literature review highlights admission control and network planning as two critical areas that require more attention as TSN systems scale and move toward wireless and hybrid deployments. Admission control must account for more than available bandwidth and instead evaluate worst-case latency, interference, and reliability before allowing new traffic flows. Network planning must similarly be treated as a first-order design concern, incorporating redundancy, synchronization strategy, and scheduling decisions early to preserve deterministic behavior across the entire system.

Several promising approaches have emerged from recent research. Improvements in time synchronization resilience, such as hot-standby clock mechanisms, help reduce the risk of timing failures. Optimization-based scheduling techniques provide more efficient ways to manage time-aware shapers, while enhanced preemption models offer finer control over critical traffic. Security-focused solutions, including intrusion detection methods that target timing-based attacks, further reinforce the need to view latency as both a performance and safety issue.

Ultimately, the future of TSN lies in hybrid architectures that combine the stability of wired networks with the flexibility of wireless systems. Achieving this will require tighter integration across protocol layers, realistic testing environments, and careful coordination between scheduling, synchronization, and security mechanisms. Continued research and practical experimentation will be essential to ensure that TSN can meet the demands of next-generation cyber-physical systems in real-world deployments.

Team Member Contributions:

Captain Russell (Team Lead): Coordinated team efforts, delegated work assignments, and contributed to the Problem Statement, Introduction, CPS networking challenges, and Conclusion sections.

John Beuter: Researched and authored the Current TSN Applications and Challenges and MAC Protocols in TSN Applications sections, focusing on the technical aspects of deterministic communication.

Nisha Raj: Contributed to the Existing Solutions, Open Problems, and Promising Solution Approaches sections, emphasizing analysis of current research and identification of future directions.

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